

Corso di Laurea Magistrale in Informatica

Evaluating automated methods for performance microbenchmark generation

Prof. Dario Di Nucci
Dott. Antonio Trovato

Ulugbek Abdimurodov
Mat.: 0522501394



Introduction and background



Performance Testing

Introduction and background



Performance Testing



Microbenchmarking

Introduction and background



ju2jmh -Augmented LLM Workflow for Benchmarking Fixed Methods in Apache Ignite

Before/After Fix Commit

1. Checkout
Parent Commit

 git checkout
<commit>

2. Build Project
with Maven

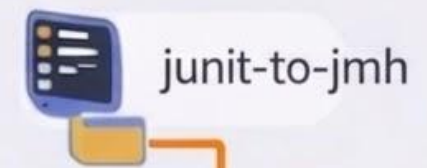
> mvn clean install
Maven

3. Generate Test Class
using GitHub Copilot

@Test
testFixedMethod()



4. Open Project
junit-to-jmh



5. Create Benchmarks
using **ju2jmh**

junit-to-jmh → ju2jmh
JUnit Test → JMH Benchmark

6. Build Ignite ju2jmh Modules

> mvn clean install
ignite-ju2jmh

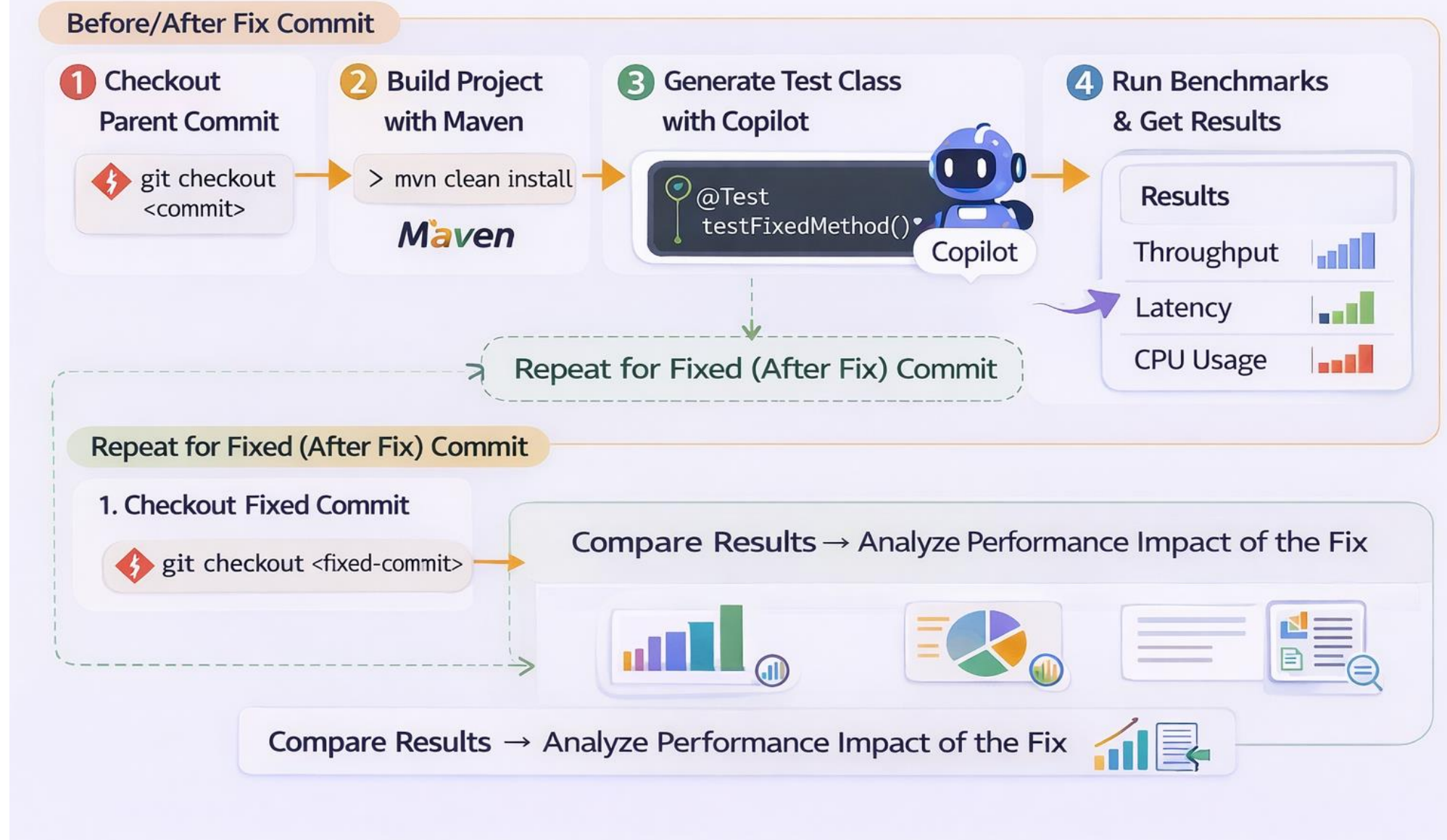
7. Run Benchmarks

jmh
java -jar benchmarks.jar

Results

Throughput 
Latency 
CPU Usage 

Standalone LLM Workflow for Benchmarking Fixed Methods





Hierarchical Bootstrap

Statistical Analysis



RQ: Between ju2jmh-augmented LLM and standalone LLM, which approach is more effective at detecting performance regressions?



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A: Standalone LLM as the primary, most effective vehicle for isolating and definitively detecting software performance regressions

Overall Detection Regression Summary

Generation Methodology	Total Benchmarks Generated	Statistically Significant Differences	Confirmed Regressions Detected	Unexpected Results	Overall Detection Rate
ju2jmh-augmented LLM	75	0	0	0	0.0%
Standalone LLM	53	6	2	4	3.8%

Why Standalone LLM won?



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- **Bypassing Functional Constraints**

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- **Bypassing Functional Constraints**
 - **Semantic understanding**
 - **Generative Flexibility**

Limitations of ju2jmh



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- **Rigidly Bound to Unit Tests**



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- **Inadequate Workload Scale**



Limitations of ju2jmh



- **Rigidly Bound to Unit Tests**
- **Inadequate Workload Scale**
- **Lack of Performance Awareness**



Extend to other systems

Test alternative LLMs

Automated Validation Loops



**Thank you for your
attention**